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$$\int_{-\infty}^{+\infty} \frac{x}{(x^2+4)^2} dx = \lim_{a \rightarrow -\infty} \int_a^0 \frac{x}{(x^2+4)^2} dx + \lim_{b \rightarrow +\infty} \int_0^b \frac{x}{(x^2+4)^2} dx$$
$$\int \frac{x}{(x^2+4)^2} dx = \frac{1}{2} \int \frac{du}{u^2}$$

$$u = x^2 + 4$$

$$du = 2x dx$$

$$\frac{1}{2} du = x dx$$

$$\Rightarrow \int \frac{x}{(x^2+4)^2} dx = \frac{1}{2} \int \frac{du}{u^2} = \frac{-1}{2u} + C = \frac{-1}{2(x^2+4)} + C$$

$$\int_{-\infty}^{+\infty} \frac{x}{(x^2+4)^2} dx = \lim_{a \rightarrow -\infty} \left[\frac{-1}{2(x^2+4)} \right]_a^0 + \lim_{b \rightarrow +\infty} \left[\frac{-1}{2(x^2+4)} \right]_0^b$$
$$= \lim_{a \rightarrow -\infty} \left(\frac{-1}{8} + \frac{1}{2(a^2+4)} \right) + \lim_{b \rightarrow +\infty} \left(\frac{-1}{2(b^2+4)} + \frac{1}{8} \right)$$
$$= \frac{-1}{8} + 0 + 0 + \frac{1}{8} = 0$$

References